Real Estate Estimator

BRAUDE College of Engineering, karmiel

Real Estate Price Evaluation Using Machine Learning

Supervisor: Dr. Reuven Cohen

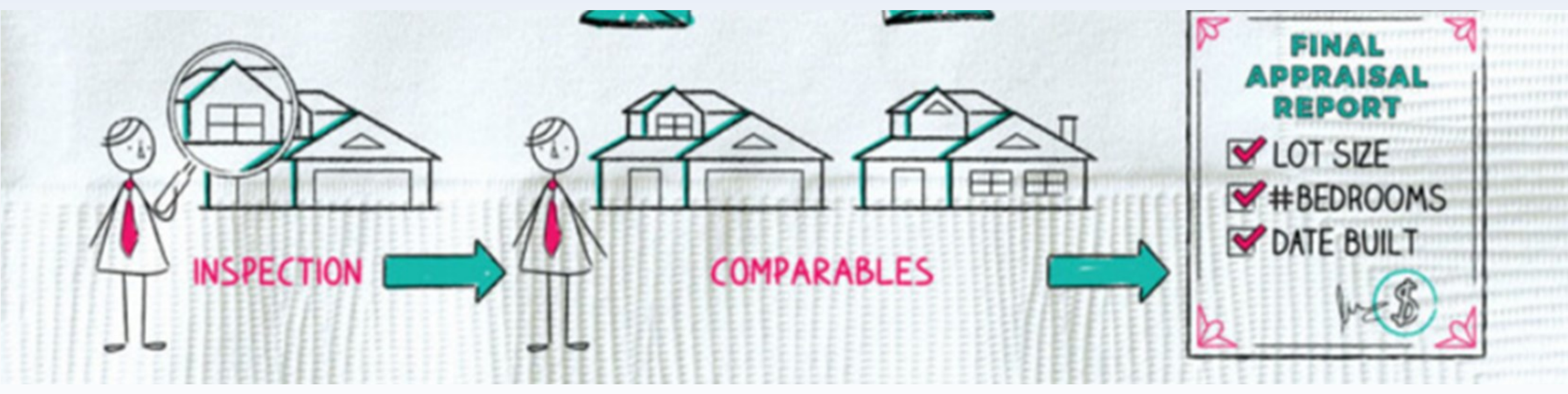
Presenting : Asaad Sajim, Joul Hourany

The Problem :

Challenges in Property Valuation:

- Difficulty Finding Comparable Properties
- Time-consuming processes.
- High Costs

Impact: Difficulty in making fair decisions for buyers and sellers.

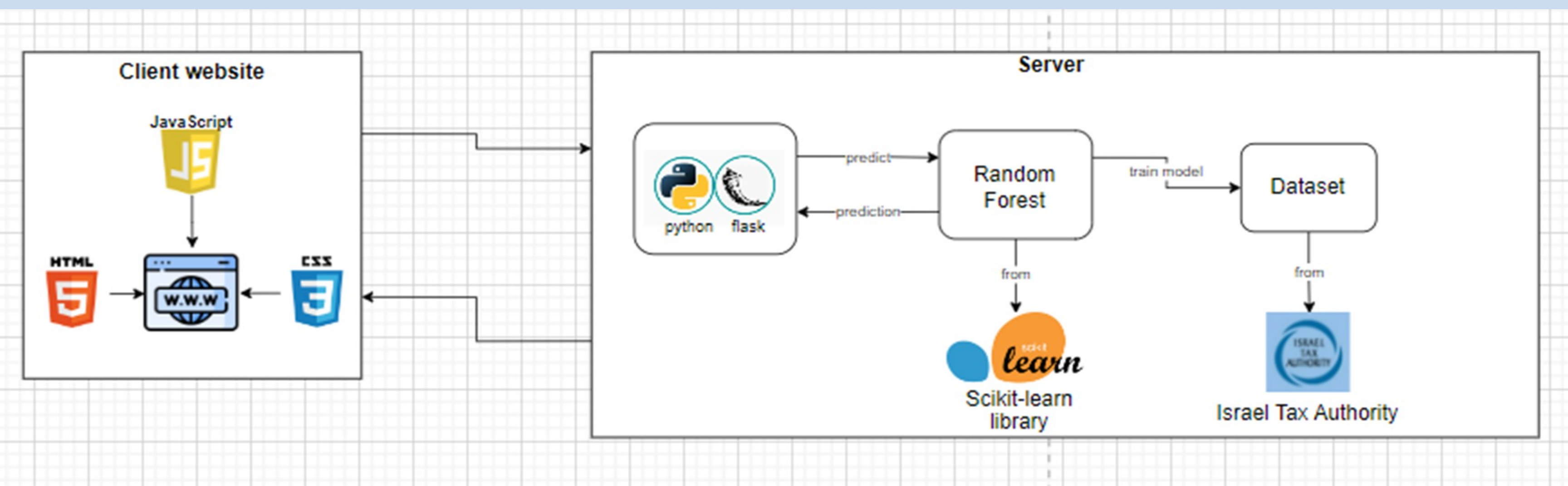


Project Goal :

A web platform that predicts real estate prices for the Israeli market.

Technologies used:

- **Machine Learning Model:** Random Forest model taken from Scikit-learn library
- **Model Training:** Trained on 2,179 property records (2024–2025) from the cities Nahariya, Karmiel, and Haifa.
- **Frontend:** React for intuitive usage and visualizations
- **Backend:** Python and Flask for processing and data analysis and results extraction.
- **Dataset:** Official real estate transaction data from the Israel Tax Authority



Future Work:

- Add new cities
- Integrate directly with the Israel Tax Authority's real estate data to automatically fetch new transaction records.
- Retrain the model regularly using the most recent data, ensuring that price estimates reflect current market trends.

System Overview :

The system is available online at **real-estate-estimator.vercel.app**

User Input	The user fills out a short form with essential property attributes Area, rooms, year built, city, block, property type, sale date, sold portion.
Data Processing	The backend validates and transforms the input. It computes features like property age and room density.
ML Prediction	The cleaned data is sent to a trained Random Forest model. The model analyzes patterns from past transactions to estimate price.
Price Range	The system returns a predicted price range. Displayed as: Estimated Price ± 8% to reflect market variability

Result

0.84

Accuracy Achieved calculated with R^2 Score

Challenges And Solutions:

- **Challenge :** Block and Parcel Number field was too specific, fragmenting the data.
  - **Solution:** Used only the first four digits of the block number to group nearby properties with similar pricing trends.

- **Challenge :** More features did not always improve performance, sometimes they added noise.
  - **Solution:** Focused on clean, reliable features (e.g., area, rooms, age) and engineered new ones like room density for better predictive accuracy.

- **Challenge :** Single predicted price did not reflect natural uncertainty in real estate markets.
  - **Solution:** Introduced a ±8% range around the predicted price to account for fluctuations and provide a more realistic price estimate.